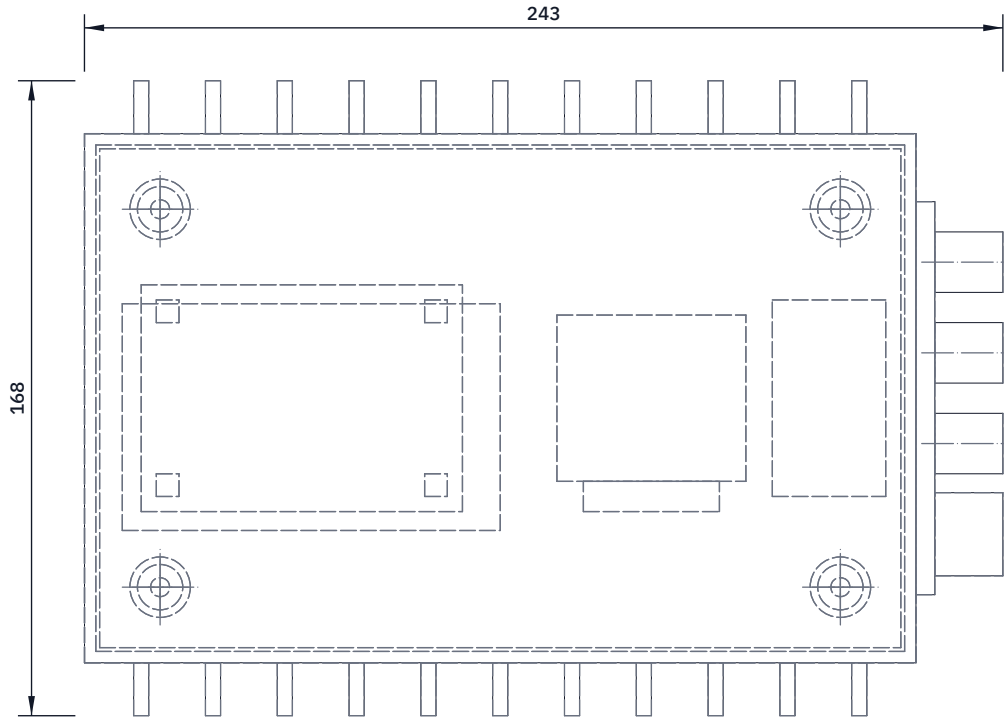


CONCEPT, NOT FOR FABRICATION

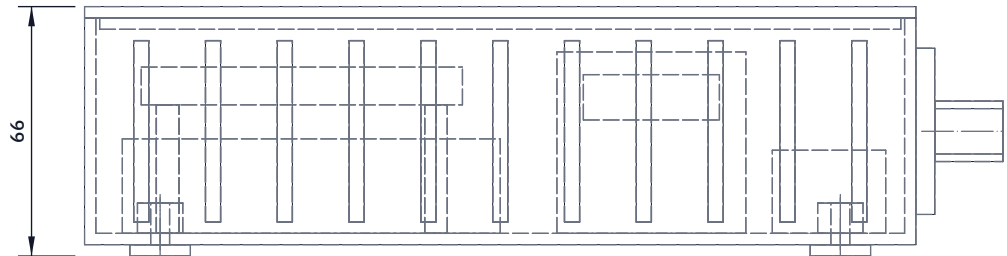
MotionCore · github.com/BoujeeEnjinia1701/motioncore

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA, interface v0.1 (MTC-CAL-001)	2026-09-25	AC
P2	Notes: 20 to 60 V, 2.0 kg limit, XT90 provisional (DDR-002)	2026-09-25	AC



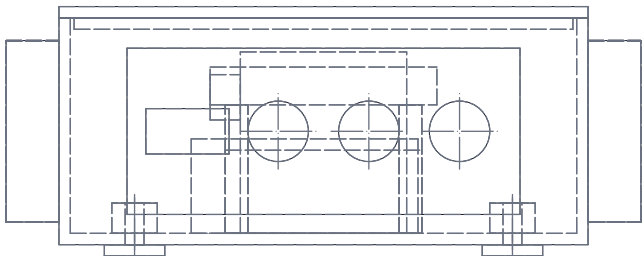
TOP VIEW

Scale 1:2; looking down (along -Z)



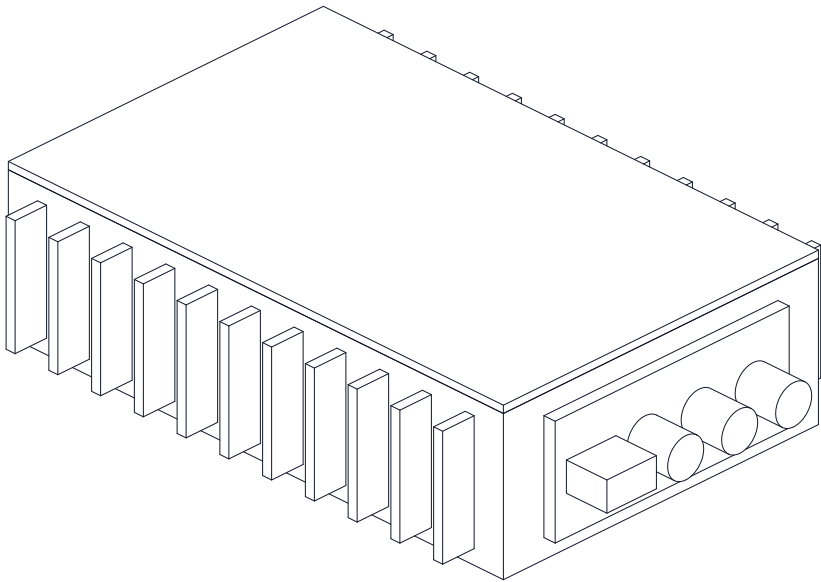
FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Overall 243 x 168 x 66 incl. pads and connectors

Body 220 x 140 x 60, wall 3, lid 3

Fins 11 per side, 4 x 14 x 48, pitch 19

Mounting: 4 x M6 on 180 x 100, pads D16 x 3

+X panel, from -Y: XT90 power in (provisional),

9-pin motor plug, M12 8-pin A safety loop,

M12 5-pin B command and CAN (250 kbit/s)

Input 20 to 60 V DC; fuse 20 A (36, 48 V) or 40 A (24 V)

Contactor 100 A, breaks 50 A at 60 V DC, 50 ms max

Precharge 100 ohm, close at 0.6 s (MTC-CAL-001)

Module about 1.94 kg (R12 limit 2.0 kg)

PRELIMINARY, NOT FOR FABRICATION

MOTIONCORE

github.com/BoujeeEnjinia1701/motioncore

General arrangement, drive and safety module

DRAWING NO. MTC-DWG-001			REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-09-25		DRAWN Amish Chadha
MATERIAL / NOTES Body and lid: extruded Al, clear anodized. Parts per bom/bom.csv					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com				Generated from cad/src; do not edit by hand	